



Cadence IPs for 2.5G and Gigabit Ethernet MAC

Part Numbers: IP701x

IP Rev: R1p12 - Doc Rev: 1.0

Integration Guide

November 6, 2017

CADENCE CONFIDENTIAL

© 2017 Cadence Design Systems, Inc. All rights reserved.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Trademarks: Trademarks and service marks of Cadence Design Systems, Inc. contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence's trademarks, contact the corporate legal department at the address shown above or call 800.862.4522. All other trademarks are the property of their respective holders.

Restricted Permission: This document is protected by copyright law and international treaties and contains trade secrets and proprietary information owned by Cadence. Unauthorized reproduction or distribution of this document, or any portion of it, may result in civil and criminal penalties. Except as specified in this permission statement, this document may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way, without prior written permission from Cadence. This document contains the proprietary and confidential information of Cadence or its licensors, and is supplied subject to, and may be used only in accordance with, a written agreement between Cadence and its customer.

Unless otherwise agreed to by Cadence in writing, this statement grants Cadence customers permission to print one (1) hard copy of this document subject to the following conditions:

1. This document may not be modified in any way.
2. Any authorized copy of this document or portion thereof must include all original copyright, trademark, and other proprietary notices and this permission statement.
3. The information contained in this document cannot be used in the development of like products or software, whether for internal or external use, and shall not be used for the benefit of any other party, whether or not for consideration.

Disclaimer: INFORMATION IN THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE AND DOES NOT REPRESENT A COMMITMENT ON THE PART OF CADENCE. EXCEPT AS MAY BE EXPLICITLY SET FORTH IN A WRITTEN AGREEMENT BETWEEN CADENCE AND ITS CUSTOMER, CADENCE DOES NOT MAKE, AND EXPRESSLY DISCLAIMS, ANY REPRESENTATIONS OR WARRANTIES AS TO THE COMPLETENESS, ACCURACY OR USEFULNESS OF THE INFORMATION CONTAINED IN THIS DOCUMENT. CADENCE DOES NOT WARRANT THAT USE OF SUCH INFORMATION WILL NOT INFRINGE ANY THIRD PARTY RIGHTS, AND CADENCE DISCLAIMS ALL IMPLIED WARRANTIES, INCLUDING MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. CADENCE DOES NOT ASSUME ANY LIABILITY FOR DAMAGES OR COSTS OF ANY KIND THAT MAY RESULT FROM USE OF SUCH INFORMATION. CADENCE CUSTOMER HAS COMPLETE CONTROL AND FINAL DECISION-MAKING AUTHORITY OVER ALL ASPECTS OF THE DEVELOPMENT, MANUFACTURE, SALE AND USE OF CUSTOMER'S PRODUCT, INCLUDING, BUT NOT LIMITED TO, ALL DECISIONS WITH REGARD TO DESIGN, PRODUCTION, TESTING, ASSEMBLY, QUALIFICATION, CERTIFICATION, INTEGRATION OF CADENCE PRODUCTS, INSTRUCTIONS FOR USE, LABELING AND DISTRIBUTION, AND CADENCE EXPRESSLY DISAVOWS ANY RESPONSIBILITY WITH REGARD TO ANY SUCH DECISIONS REGARDING CUSTOMER'S PRODUCT.

Restricted Rights: Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

Table of Contents

1 Introduction	8
2 Design Integration	10
2.1 Delivery Directories	10
2.2 Design Configuration	10
2.2.1 Option with Perl TK	11
2.2.2 Option without Perl TK	11
2.3 Interface Integration.....	11
2.4 Integrating DFT.....	11
2.5 Integrating Technology-Specific Cells	12
2.6 Integrating RAMs.....	12
2.6.1 External Memories	12
2.6.2 Determining RAM Size.....	12
2.6.3 Reducing RAM Size.....	13
2.6.4 Dual-Port External Memory.....	13
2.6.5 Single-Port External Memory	14
2.7 Integrating with System Clocking and Reset Scheme.....	14
2.8 Integrating with System Low-Power Scheme	14
2.9 Verification Features.....	15
2.10 Diagnostic Features.....	15
2.10.1 Diagnostic Features for ASF.....	15
2.11 Compiling the Design	15
2.12 Running Lint	15
2.13 Running CDC	15
3 Synthesis	16
3.1 Directories / Files.....	16
3.2 Synthesis Constraints.....	16
3.3 Clocking Constraints.....	16
3.3.1 Combinatorial Paths.....	17
3.4 I/O Constraints.....	17
3.5 Timing Modes	17
3.6 Power Constraints	17
3.7 Handling Integrated RAMs	18
3.8 Preserving Key Parts of the Design.....	18
3.9 I/O Driving/Loading Requirements	18
3.10 Latches.....	18
3.11 Post-Synthesis Logic Equivalency Checks (LEC)	18
4 Design for Test	19
5 Performance, Power, and Area	20

List of Tables

Table 1: Part Numbers	6
Table 2: References and Related Documents	6
Table 3: Acronyms and Abbreviations	6
Table 4: Typographical Conventions	7
Table 5: Delivery File Details	10
Table 6: Example of Determining RAM Size	12
Table 7: Example of Reducing RAM Size (TX)	13
Table 8: Example of Reducing RAM Size (Partial Store and Forward Mode)	13
Table 9: Recommended Operating Frequencies	14
Table 10: Typical Controller Configurations	20
Table 11: PPA Information	20
Table 12: Example Area	21

List of Figures

Figure 1: Example System-Level Block Diagram of IP7017 and IP7017A	8
Figure 2: Example System-Level Block Diagram of IP7016	8
Figure 3: Example System-Level Block Diagram of IP7014 and IP7014A	9
Figure 4: Example System-Level Block Diagram of IP7010	9

About this Document

This document describes the integration aspects of the Cadence IPs for 2.5G and Gigabit Ethernet MAC (as listed in *Table 1*) into systems on chip (SoC). The intended audience for this document is design engineers.

This guide is organized as follows:

Section 1 Introduction on page 8

Section 2 Design Integration on page 10

Section 3 Synthesis on page 16

Section 4 Design for Test on page 19

Section 5 Performance, Power, and Area on page 20

Part Numbers

The document is applicable to the Cadence product part numbers listed in the following table:

Table 1: Part Numbers

Part Number	Product Name
IP7017	Cadence IP for 2.5G Ethernet MAC with DMA, 1588, TSN/AVB, and PCS
IP7017A	Cadence IP for 2.5G Ethernet MAC with DMA, 1588, TSN/AVB, and PCS for Automotive
IP7016	Cadence IP for 2.5G Ethernet MAC
IP7014	Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB
IP7014A	Cadence IP for Gigabit Ethernet MAC with DMA, 1588, and TSN/AVB for Automotive
IP7010	Cadence IP for Gigabit Ethernet MAC

References and Related Documents

This guide references the following documents:

Table 2: References and Related Documents

Reference	Version/Description
IEEE 802.3 standard	2015 base standard
IEEE 802.3br	Frame Pre-Emption (or Interspersing Express Traffic)
IEEE 802.1Qbv	Enhancements for Scheduled Traffic
Cadence IP for GEM User Guide	Refer to <i>Table 1</i> for the name of your IP for the User Guide.

Acronyms and Abbreviations

The following acronyms and abbreviations are used in this guide:

Table 3: Acronyms and Abbreviations

Term	Definition
AMBA	Advanced Microcontroller Bus Architecture
AHB	Advanced High-Performance Bus
APB	Advanced Peripheral Bus
ASF	Automotive Safety Feature
AVB	Audio Video Bridging

Table 3: Acronyms and Abbreviations (Continued)

Term	Definition
AXI	Advanced eXtensible Interface
BIST	Built-In Self Test
DFT	Design For Test
DMA	Direct Memory Access
DUT	Device Under Test
GEM	Short-form name for Cadence IPs for 2.5G and Gigabit Ethernet MAC
ECC	Error Correcting Code
HS/LS	High Speed/Low Speed
LEC	Logical Equivalency Check
PCS	Physical Coding Sublayer
PHY	Physical Hardware Layer
PLL	Phase Locked Loop
SoC	System on Chip
TB	Testbench
TBI	Ten-Bit Interface
TSN	Time-Sensitive Networking
TSU	Timestamp Unit

Typographical Conventions

The following conventions are used in this guide:

Table 4: Typographical Conventions

Convention	Description
<code>courier font</code>	Indicates code

1 Introduction

The Cadence IPs, as listed in *Table 1 Part Numbers* on page 6, subsequently also referred to in this document as GEM, is a highly customizable soft controller IP compliant with IEEE 802.3 standard.

Figure 1 to Figure 4 show example system-level block diagrams of GEM variants.

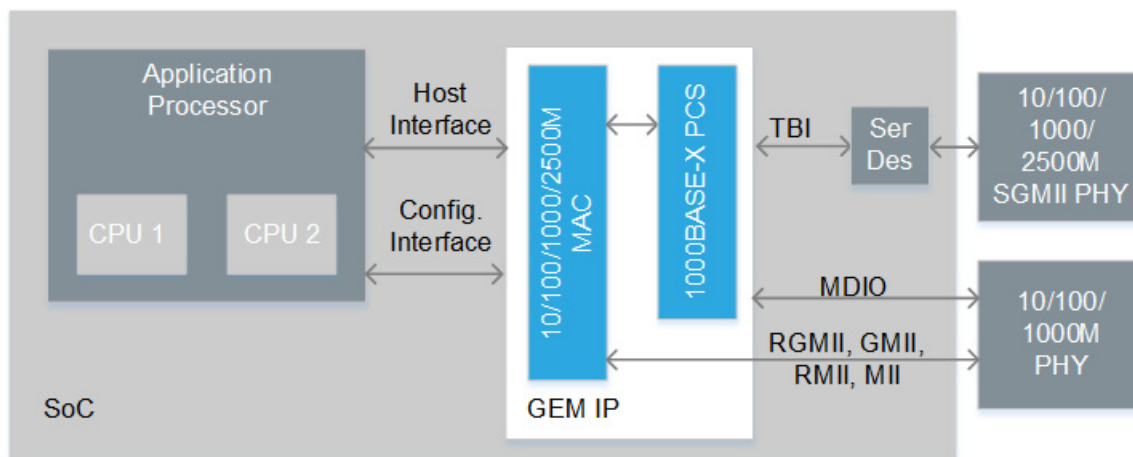


Figure 1: Example System-Level Block Diagram of IP7017 and IP7017A

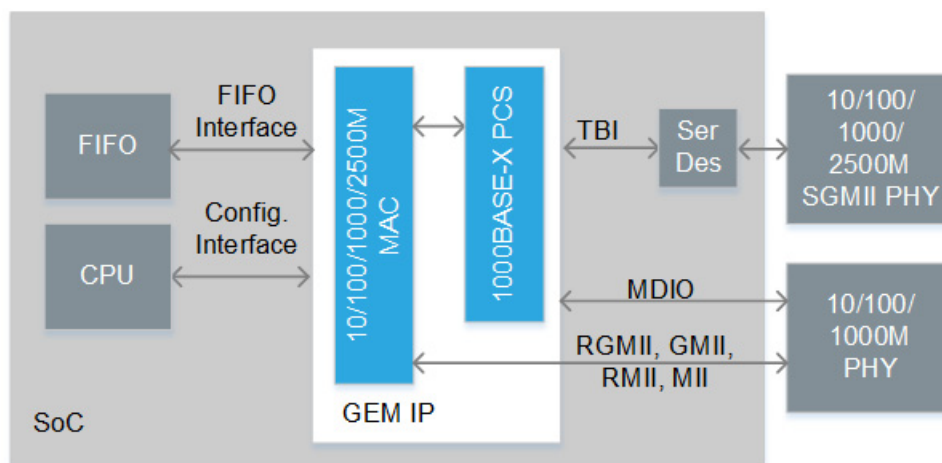


Figure 2: Example System-Level Block Diagram of IP7016

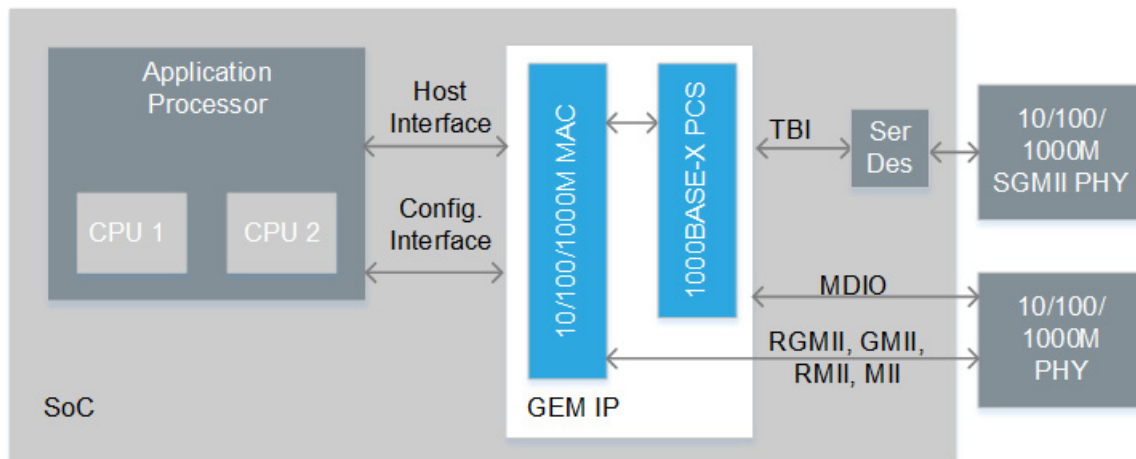


Figure 3: Example System-Level Block Diagram of IP7014 and IP7014A

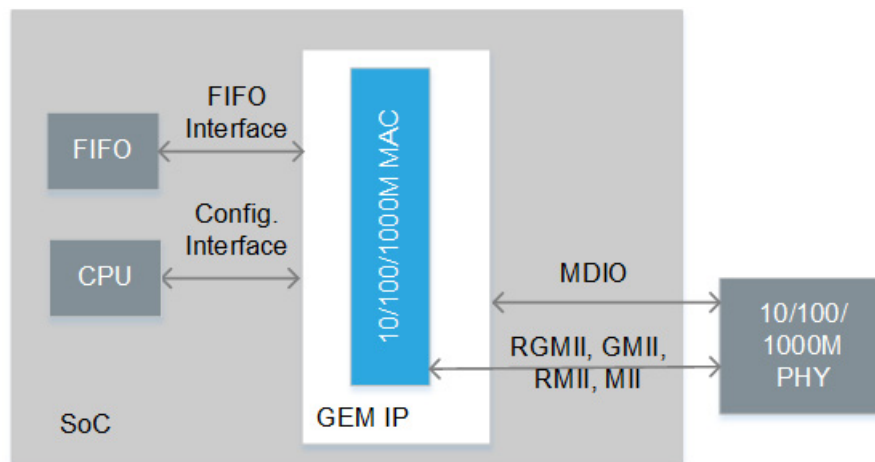


Figure 4: Example System-Level Block Diagram of IP7010

2 Design Integration

This chapter provides the information for integrating GEM into your SoC designs.

2.1 Delivery Directories

Table 5 lists the package content and short description of the deliverables. Please refer to the provided RELEASE_TREE and RELEASE_NOTES files in the base folder of the package for further information on contents and usage.

Table 5: Delivery File Details

Directory	Subdirectory/Files	Description
dft		This directory is empty for GEM
doc		Contains all relevant IP documentation, including quick start guide, user guide, integration guide, and testbench descriptions
func_ver	gatesim	Gate simulation information
	hvm	High-volume manufacturing testbench (TB)
	sanity	Sanity/two TBs (one is a pure Verilog directed TB and the other demonstrates Cadence VIP and test scenarios written in C)
	sva	SystemVerilog assertions
	sdf	Delay back-annotation files
hdl	hdl_src	Unencrypted synthesizable design source
hdl_qc		Setup files for synthesis and linting scripts
software	CoreDriver	Core driver source code
	Doc	Driver documentation
	README.txt	
	ref_code	Linux driver reference code
	RELEASENOTE.txt	Release history
synth	constraints	Functional and scan mode constraints files
	logs	This directory is empty for GEM
	technology	This directory is empty for GEM
	scripts	Support scrips for synthesis run and analysis
work	Makefile	Make file for running simulation and synthesis scripts
	gem_cfg_builder.pl	Script for creating ancillary files for a particular configuration
	cfg_gen_pkg	Directory containing data associated with previously generated configurations and utilities used by the gem_cfg_builder.pl script

2.2 Design Configuration

Before GEM can be used, you must create a design configuration matching the user's requirements. GEM has many parameterization options to configure the IP during the compilation stage. This is achieved using Verilog `define compiler directives in an include file called gem_gxl_defs.v.

To aid in debug, these configurations are readable through the APB address space starting from address 0x280 (Refer to *Section 11 Registers* of the GEM User Guide for further details).

The release is packaged with a configurator tool called gem_cfg_builder.pl to aid the user in creating configuration-specific files needed for compilation and synthesis, and also some other useful files that can help you to understand any LINT/CDC paths that might show in SoC-level quality flows. More information on this tool can be found by launching it with the -help argument.

This tool can be launched as follows:

```
cd gem_gxl/work
```

```
./gem_cfg_builder.pl -help
```

2.2.1 Option with Perl TK

The first and best option is to launch the included configuration GUI tool that requires a Perl TK package. Offering detailed descriptions of the options available, this GUI-based tool can automatically generate the configuration files needed in order to simulate and synthesize the design with the appropriate options. The main files generated are a file listing, a configuration Verilog defines file, and a synthesis constraints file.

The tool can be launched from the work directory as follows:

```
cd gem_gxl/work  
  
./gem_cfg_builder.pl -gui
```

This tool can save a configuration to a file with a <user_config_name>.cee extension. It is recommended the configuration is saved for future reference. Also if this file is provided back to Cadence, some configuration-specific register views can be generated such as IP-XACT, RDL, and register descriptions in RTF format.

2.2.2 Option without Perl TK

If Perl TK is not available, the second option is to create the Verilog defines file manually, and then use the supplied `cfg_builder.pl` tool to generate the remaining configuration-specific files. The delivered database is provided with a few example fixed configurations, such as:

```
gem_gxl/work/cfg_gen_pkg/fixed_cfgs/pbuf_3qs_axi.cee.
```

This is a AXI multi-queue DMA packet buffer configuration.

1. First use the `cfg_builder` tool to generate the Verilog defines file for this example configuration

```
cd gem_gxl/work  
  
gem_cfg_builder.pl -cfg pbuf_3qs_axi
```

2. Open “`gem_gxl_defs.v`” file generated in step 1 using a text editor and follow the comments for each option update to the desired configuration. The comments should provide sufficient information to aid selection.
3. Save the updated Verilog defines file as <user_config_name>.v
4. Import the updated defines file to generate a configuration file in the .cee extension format

```
gem_cfg_builder.pl -import <user_config_name>.v
```

5. The <user_config_name>.cee will now be stored in `./cfg_gen_pkg/imported_cfgs`
6. Run the config builder tool again to generate some important files for simulation and synthesis.

```
gem_cfg_builder.pl -cfg <user_config_name> -gen_synth
```

2.3 Interface Integration

The AMBA interfaces fully conform to the ARM AMBA Revision 2.0 specification.

2.4 Integrating DFT

This design does not require integration of Design For Test (DFT) features.

2.5 Integrating Technology-Specific Cells

The design is implemented as fully synthesizable RTL. All synchronizers are implemented using a module called `cdnsdru_datasync_v1`. You may replace this module using a technology-specific cell should you wish to do so.

2.6 Integrating RAMs

This section is relevant only for GEM IPs with DMA capabilities (i.e. IP7017A, IP7017, IP7014, and IP7014A); and only when they have been configured to have the packet buffer option.

2.6.1 External Memories

Two additional memories are required to be added externally to GEM to provide the packet buffering. These must be connected to the transmitter and receiver packet buffer interfaces at the top level. The external memory type can either be dual port or single port.

The packet buffer implementation works with 32, 64, or 128-bit AMBA databus widths.

In the transmit direction, the DMA will continue to fetch packet data up to a limit of 256 packets, or until either the external buffer is full. The size of the external buffer may be programmed to use either all or half the address range as specified in the configuration file. In the default configuration, the buffer address width is set to 10 bits, although the actual size required by the user can be set via the design configuration file. In 32-bit mode, a setting of 10 bits relates to a maximum useable size of 4KB. Similarly, in 64-bit mode it relates to a size of 8KB.

In the receive direction, if the external buffer becomes full, then an overflow will occur. An overflow will also occur if the limit of 256 packets is breached. The size of the external buffer may be programmed to use either all, half, quarter, or an eighth of the address range as specified in the configuration file. In the default configuration, the buffer address width is set to 11 bits. In 32-bit mode, this relates to a maximum external buffer size of 8KB. Similarly, in 64-bit mode it relates to a size of 16KB. Again, these sizes may be scaled during configuration.

2.6.2 Determining RAM Size

To guarantee maximum bandwidth in all modes of operation, the general rule for the RAM is a size equal to twice the anticipated maximum frame size. Having a RAM that is twice the maximum frame size allows one frame to be forwarded while the next frame is being written. For example, in the transmit direction one frame will be transmitted through the MAC while another frame is being read from the host AMBA interface. This configuration ensures maximum throughput when maximum sized frames are transmitted / received consecutively, as both halves of the DMA can operate concurrently.

For transmit, each priority queue has an independent reserved area of RAM, and the above rule should be met on each queue's RAM segment – i.e. when priority queuing mode and full store and forward mode is enabled, every configured region of the attached memory attributed to a priority queue must be greater than 2X the maximum transmitted frame length. For receive, all queues share the single RAM space and the above rule need only be applied to the full RAM. *Table 6* shows an example of using a 4 queue configuration and a maximum sized frame of 15KB to guarantee maximum performance in all modes of operation. The example shows a transmit path configured to have four segments of equal size.

Table 6: Example of Determining RAM Size

Direction	RAM Sizes (KB)	Queue Number	RAM Area of the Queue (KB)
Transmit	128	3	32
		2	32
		1	32
		0	32
Receive	32	N/A	32

2.6.3 Reducing RAM Size

The RAM sizes described above are cautious and are not always necessary. If maximum sized frames are not transmitted consecutively, then a smaller RAM size is permitted. In full store and forward mode, the minimum RAM size for correct functional operation must be greater than the max frame length. Also, if the core is configured in full duplex operation, the RAM size on the transmit side can be reduced as detailed below.

2.6.3.1 Transmit Operation

The transmit DMA design is efficient and recovers memory as frames are transmitted through the MAC. After each byte is transmitted the memory area used by this byte will be recovered. By recovering memory locations as frames are transmitted then space is available for the next frame and this frame can be read over the AMBA bus. If the maximum frame size is less than 80% of the allocated transmit RAM size reserved per queue, and the average AXI or AHB bandwidth is significantly higher than the required line rate, then it is unlikely that there will be a performance impact. The 20% overhead allows the DMA to maintain overall line rate. **Note:** This operation does not apply in half duplex mode as the frame may need retransmitted owing to a collision and the memory used by the transmit frame is only cleared after the frame has fully transferred through the MAC. *Table 7* shows the memory size for transmitting a maximum frame length of for example 10KB could be transmitted and received at line rate with the following SRAM sizes.

Table 7: Example of Reducing RAM Size (TX)

Direction	RAM Sizes (KB)	Queue Number	RAM Area of the Queue (KB)
Transmit	64	3	16
		2	16
		1	16
		0	16
Receive	32	N/A	32

2.6.3.2 Partial Store and Forward Mode

Partial store and forward mode allows frames that are larger than the RAM size to transfer through the design. If under normal circumstances, the maximum sized frame is 2KB. But frames larger than this size may occasionally occur, then partial store and forward mode could be activated with a threshold of 2KB. *Table 8* shows the scenario where the buffer sizes could be reduced.

Table 8: Example of Reducing RAM Size (Partial Store and Forward Mode)

Direction	RAM Sizes (KB)	Queue Number	RAM Area of the Queue (KB)
Transmit	16	3	4
		2	4
		1	4
		0	4
Receive	4	N/A	4

When 2KB of data is has been written to memory then it will be forwarded, regardless of whether or not the full frame has been written. The memory used by this initial 2KB will be recovered as the frame is forwarded, which allows the remainder of the frame to be written. With partial store and forward mode, there is the risk of underflow (transmit path) or overflow (receive path) if the AMBA interface does not keep up with the frame rate. This risk is however low if the AMBA rate is faster than the MAC rate – in most systems, it is typically faster and this underflow/overflow risk is low. **Note:** if partial store and forward mode is to be used, then the define 'pbuf_cutthru' should be set in the configuration file.

2.6.4 Dual-Port External Memory

Note: Although these memories must have 2 ports, each port has only a single access direction, i.e. port A of the TX DPRAM is WRITE only, while port B is READ only.

Dual port external memory has no frequency limitations on aclk/hclk and these clocks can therefore run faster or lower than the MAC clocks (rx_clk and tx_clk).

The external DPRAM modules that must allow fully independent read and write interfaces synchronous to different clock domains and using independent addressing. The packet buffer implementation will ensure that the same memory address is not accessed simultaneously by both the read and write ports. This is effectively an underflow condition of the buffer which is not possible.

The connections to the DPRAM modules are as follows:

- The write port of the transmitter DPRAM must be connected to the txdpram_*a signals, all of which are AMBA hclk (or aclk for AXI) timed and hence this port must be clocked by hclk/aclk
- The read port of the transmitter DPRAM must be connected to the txdpram_*b signals, all of which are MAC tx_clk timed and hence this port must be clocked by tx_clk
- The write port of the receiver DPRAM must be connected to the rxdpram_*a signals, all of which are MAC rx_clk timed and hence this port must be clocked by rx_clk
- The read port of the receiver DPRAM must be connected to the rxdpram_*b signals, all of which are AMBA hclk timed and hence this port must be clocked by hclk/aclk

2.6.5 Single-Port External Memory

GEM_GXL can be configured to optionally use an external single port memory with a 128 bit data bus. For this mode of operation all memory signals are synchronous to aclk/hclk and both the receive and transmit memory must be therefore be connected to aclk/hclk.

Single port external memory operation however has some frequency limitations, where the aclk/hclk frequency must be faster than the MAC data rate. *Table 9* details recommended minimum operating frequencies for all possible configurations and MAC speeds:

Table 9: Recommended Operating Frequencies

DMA Bus Width	MAC Rate (Mbps)	Minimum AHB/AXI Frequency (MHz)
32	1000	125
32	100	15
32	10	10
64	1000	65
64	100	10
64	10	10
128	Not a supported configuration with a single-port external memory configuration	Not applicable

GEM does not have automatic built-in support for powering down external memory when using power saving modes of operation.

2.7 Integrating with System Clocking and Reset Scheme

Refer to *Section 9.2 Clock Requirements* in the GEM User Guide.

Resets are required to be asserted asynchronously and deasserted synchronously to their relevant clock domains.

2.8 Integrating with System Low-Power Scheme

This design does not include low-power features.

2.9 Verification Features

Setting the `retry_text` bit aids directed functional testing by speeding up the collision back-off algorithm, the pause counter, and the frer timeout.

2.10 Diagnostic Features

Setting the `loopback_local` bit in the network control register asserts the `loopback_local` signal to the system clock generator. This also connects `txd` to `rx_dv` and `tx_en` to `rx_dv` as well as forces full-duplex mode. If the GEM configuration contains the PCS, then bit 11 of the network configuration register must be set low to disable TBI mode when in internal loopback. Local loopback functionality is a configuration option of GEM

2.10.1 Diagnostic Features for ASF

This subsection is relevant only for GEM IP7014A.

Fault status is reported in the following Automotive Safety Feature (ASF) registers:

- `asf_sram_corr_fault_status`
- `asf_sram_uncorr_fault_status`
- `asf_sram_fault_stats`
- `asf_trans_to_fault_status`
- `asf_protocol_fault_status`

Refer to the GEM User Guide for more details.

2.11 Compiling the Design

The configurator tool (see *Section 2.2 Design Configuration* on page 10) generates the `gem_gxl.f` which lists the files for compilation.

2.12 Running Lint

The configurator tool (see *Section 2.2 Design Configuration* on page 10) generates the waiver files for AFL and Spyglass.

2.13 Running CDC

The configurator tool (see *Section 2.2 Design Configuration* on page 10) generates the waiver files for Cadence Conformal[®] Constraint Designer.

3 Synthesis

Example synthesis scripts are provided with the design to demonstrate how to constrain the clocks and I/O of GEM.

Synthesis constraint files are generated with the `gem_cfg_builder.pl` script. The scripts for synthesis are gathered in the `gem_gxl/synth/scripts` directory and its subdirectories.

3.1 Directories / Files

Below are the main subdirectories for performing synthesis:

```
`-- synth
    |-- constraints
    |-- logs
    |-- reports
    |-- scripts
    |   |-- cpf
    |   |-- rc
    |   |-- setup
`-- hdl_qc
    |-- cfg
    |   |-- <user_config_name>
```

- Synthesis scripts are provided in `gem_gxl/synth/scripts`
- Setup files are provided in `gem_gxl/hdl_qc/cfg/<user_config_name>`

Synthesis is run using `make` in the `gem_gxl/work` directory.

3.2 Synthesis Constraints

To run a trial synthesis, first run the config builder tool to generate relevant files for synthesis:

```
gem_cfg_builder.pl -cfg <user_config_name> -gen_synth
```

The SDC constraints will be placed in:

```
gem_gxl/synth/constraints/cfg/<user_config_name>/gem_<user_config_name>.func.sdc
```

Setup files, such as `project.tcl` and `setup_project.csh`, needed by the flow are also generated and placed in the directory:

```
gem_gxl/hdl_qc/cfg/<user_config_name>/
```

Please note that the scripts require access to technology libraries referenced by

```
gem_gxl/synth/scripts/tech_lib_setup.tcl
```

An example of the files needed are shown in:

```
gem_gxl/synth/scripts/tech_lib_example.tcl
```

3.3 Clocking Constraints

Signals passing between clock domains are controlled by handshaking logic. Even though there is no combinatorial logic between two flip-flops in different clock domains, this handshaking logic requires that the maximum delay path between the two clock domains is constrained to one clock period of the destination clock.

The receive path interface between the `gem_pcs` and `gem_rx` passes data between `rbc1` or `pcs_rx_clk` and `rx_clk` clocks. In gigabit TBI mode, the `rx_clk` should be sourced from `rbc1` or `pcs_rx_clk` which requires that these two clock trees are balanced with each other.

Similarly, the transmit path interface between the `gem_pcs` and `gem_tx` passes data between `gtx_clk` and `tx_clk` clocks. In gigabit TBI mode, the `tx_clk` should be sourced from `gtx_clk` which requires that these two clock trees are balanced with each other.

RGMII operation makes use of inverted transmit and receive clocks. These clocks need to be balanced with their inverted versions.

RMII operation makes use of `rmii_tx_clk` and `rmii_rx_clk` generated from `ref_clk`. Care must be taken to ensure that these clocks are driven correctly with respect to `ref_clk` to ensure adequate setup and hold time for signals passing to and from the 50MHz `ref_clk` domain. Balancing `tx_clk` with `ref_clk` will ensure that these setup and hold times are met on the transmit path. On the RMII receive path, there is 20ns of slack on the receive signals passing from the RMII module to the MAC receive module, so balancing `rx_clk` with `ref_clk` should not be required.

3.3.1 Combinatorial Paths

When configured for RGMII operation, there are two combinatorial paths from `rgmii_tx_clk_sig` and ending at `rgmii_txd` and `rgmii_tx_ctl`.

`rgmii_tx_clk_sig` is sourced from `rgmii_tx_clk` and is used to control the multiplexor driving `rgmii_txd` and `rgmii_tx_ctl` which switches every 4ns.

The delay of `rgmii_tx_clk_sig` relative to `rgmii_tx_clk` should be equivalent to the clock to output delay of the flip-flops feeding the multiplexor.

The min and max delays of `rgmii_tx_clk_sig` to the `rgmii_txd` and `rgmii_tx_ctl` outputs should be such that changes `rgmii_txd` and `rgmii_tx_ctl` occur coincidentally and appropriately offset from the `tx_clk` output.

The RGMII `tx_clk` output requirements are described in the RGMII clocking the Clocks and Resets chapter of the GEM User Guide.

3.4 I/O Constraints

I/O constraints are defined in the generated constraints file. These are set to be 0.6 of the relevant clock period. For example:

```
# Clock rx_clk, constrained to 125Mhz
set rx_clk_period [expr 8 * $clk_overconstraint]
set rx_clk_generic_indel [expr $rx_clk_period * 0.6]
set rx_clk_generic_outdel [expr $rx_clk_period * 0.6]
```

3.5 Timing Modes

The generated constraints file support a single timing mode.

3.6 Power Constraints

Power constraints do not apply to this IP.

3.7 Handling Integrated RAMs

The RAMs are not integrated into the IP but connected through top-level I/Os. Refer to *Section 2.6 Integrating RAMs* on page 12 for more details.

3.8 Preserving Key Parts of the Design

This section is relevant only for GEM IP7014A.

Automotive Safety Features (ASF) include options to duplicate the Timestamp Unit (TSU) and the transmit scheduler. If these options are selected the user should check that the duplicated functionality is not optimized out during their logic synthesis flow.

3.9 I/O Driving/Loading Requirements

In the generated constraints file, the cell driving the inputs is defined to be: \$DRIVE_CELL. This variable needs to be initialized in the setup TCL file.

The loads on the output signals are defined to be either five or twenty times 2.3fF depending on whether they are internal or external drivers:

```
# Set the load on the outputs
```

```
set INVCAP 0.0023
```

```
set_load [expr "5 * $INVCAP"] [get_ports $INT_DRIVERS]
```

```
set_load [expr "20 * $INVCAP"] [get_ports $EXT_DRIVERS]
```

3.10 Latches

There are no latches in the design.

3.11 Post-Synthesis Logic Equivalency Checks (LEC)

After completing synthesis in the gem_gxl/work directory, LEC is run with the following commands:

```
cd ../work_phys_<config_name>
../synth/scripts/run_phys.csh -lec_2stage
```

This performs Logic Equivalency Checking for rtl2mapped and map2final. The results for the rtl2mapped and map2final runs can be respectively found in:

synth/default/reports_<config>/rtl2mapped.lec.log and synth/default/reports_<config>/map2final.lec.log

4 Design for Test

Other than the `retry_test` bit in the network configuration register, the IP core contains no special DFT features. Setting the `retry_test` bit aids directed functional testing by speeding up the collision back-off algorithm, the pause counter decrement, and the frer timeout.

The flip-flops in the IP core are fully scannable with all clocks brought out at the top-level of the IP core. Trial ATPG has been run with the `pbuf_3qs_axi` configuration and achieved a stuck-at coverage of 99.82 % and at-speed coverage of 89.97 %.

5 Performance, Power, and Area

Power consumption depends on the final configuration and use case (clock frequency, data transmission parameters, etc.). As such, the figures presented in this section should be considered as informative only. Actual obtained figures will vary depending on the chosen technology and physical implementation.

Table 10 shows the controller configurations used for obtaining the physical estimates listed in Table 11.

Table 10: Typical Controller Configurations

IP7017	IP7017A	IP7016
• Basic configuration • DMA (AXI) option • GEM statistic registers • Three priority queues • Four screener registers • PCS functionality • AXI frequency 400MHz • APB frequency 100MHz • AXI bus width 64 bits • Ethernet speed 2.5Gbps	• Basic configuration • DMA (AXI) option • GEM statistic registers • Three priority queues • Four screener registers • PCS functionality • AXI frequency 400MHz • APB frequency 100MHz • AXI bus width 64 bits • Ethernet speed 2.5Gbps • Main ASF features with SRAM ECC and TSU duplication	• Basic configuration • GEM statistic registers • Four priority queues • Four screener registers • PCS functionality • APB frequency 100MHz • FIFO bus width 64 bits • Ethernet speed 2.5Gbps
IP7014	IP7014A	IP7010
• Basic configuration • DMA (AXI) option • GEM statistic registers • Three priority queues • Four screener registers • PCS functionality • AXI frequency 400MHz • APB frequency 100MHz • AXI bus width 64 bits • Ethernet speed 1Gbps	• Basic configuration • DMA (AXI) option • GEM statistic registers • Three priority queues • Four screener registers • PCS functionality • AXI frequency 400MHz • APB frequency 100MHz • AXI bus width 64 bits • Ethernet speed 1Gbps • Main ASF features with SRAM ECC and TSU duplication	• Basic configuration • GEM statistic registers • Four priority queues • Four screener registers • PCS functionality • APB frequency 100MHz • FIFO bus width 64 bits • Ethernet speed 1Gbps

Table 11: PPA Information

Foundry Process		Industry standard 28nm process and standard cell library					
Power Consumption (mW)							
		IP7017	IP7017A	IP7016	IP7014	IP7014A	IP7010
Running at the above frequencies at maximum data transfer rates	Leakage	0.47	0.74	0.07	0.42	0.64	0.06
	Internal	8.17	8.95	2.48	4.80	5.28	0.94
	Net	1.30	1.59	0.36	0.72	0.78	0.13
	Total	9.94	11.28	2.91	5.94	6.70	1.13
Area (k gates)							
		IP7017	IP7017A	IP7016	IP7014	IP7014A	IP7010
NAND2X1 Equivalent Gate Count ¹		414	534	81	400	516	71

¹NAND2X1 gate count equivalent is calculated based on the following:

- Scan structures excluded

- Uses smallest NAND2X1 drive strength in target library – a gate size of 0.3645 μm^2 was used to calculate the gate count number

Note (Table 11):

- These are typical power numbers
- The area values depends on the configuration you choose
- The area value does not include the RAM required by certain configurations
- These numbers were calculated after running a post-layout gate-level simulation with appropriate net toggling information.

Table 12 shows the approximate area required by various options of GEM.

Table 12: Example Area

Feature	Approx. Area ² Added (k)	Applicable for					
		IP7017	IP7017A	IP7016	IP7014	IP7014A	IP7010
Basic configuration	49	Y	Y	Y	Y	Y	Y
PCS option	9	Y	Y	Y	Y	Y	N
TSU option	23	Y	Y	N	Y	Y	N
GEM statistics registers option	22	Y	Y	Y	Y	Y	Y
DMA packet buffer (AHB) option	72	Y	Y	N	Y	Y	N
DMA packet buffer (AXI) option - single queue	88	Y	Y	N	Y	Y	N
-- each additional queue adds	16	Y	Y	N	Y	Y	N
-- with max size descriptor buffers adds (per Q)	46	Y	Y	N	Y	Y	N
802.1Qbv	11	Y	Y	N	Y	Y	N
Single-port SRAM support	26	Y	Y	N	Y	Y	N
LSO adds	9	Y	Y	N	Y	Y	N
-- LSO adds for each additional queue	4	Y	Y	N	Y	Y	N
RSC (2 queues)	57	Y	Y	N	Y	Y	N
RSC (4 queues)	96	Y	Y	N	Y	Y	N
802.3br (FIFO interface configuration)	77	Y	Y	N	Y	Y	N
802.3br (AXI DMA configuration)	195	Y	Y	N	Y	Y	N
Main ASF features	40	N	Y	N	N	Y	N
ECC protection (1 Queue)	5	N	Y	N	N	Y	N
TSU duplication	38	N	Y	N	N	Y	N

²Approximate two input NAND equivalent gate counts

Note (Table 12):

- The numbers reported are based on a synthesis and trial place and route flow using a TSMC 28nm HPM process
- Gate sizes of 0.3645 μm^2 were used to calculate the specific gate count number
- This area estimate may significantly change based on gate size, relative size of flip-flops, and technology used

Change Log

Revision	Date	Description
1.0	November 6, 2017	Initial version with new content as well as content from the legacy user guide doc # I-IPA01-0266-USR, revision 18 (22 June 2017)